

Case Study on the Analysis of SDLC Models Applied to an Instagram Software Application

Aim

To analyze the Software Development Life Cycle (SDLC) models used in the development of the Instagram software application and identify the most suitable SDLC model.

Objective

- Study different SDLC models.
- Analyze the development process of the Instagram application.
- Compare the advantages and limitations of various SDLC models.
- Recommend the most appropriate SDLC model for Instagram.

Selected Software Application

Instagram – Social Media Platform

Instagram allows users to register/login, upload photos and videos, share stories and reels, send direct messages, follow users, like and comment on posts, and receive personalized content recommendations.

Functional Requirements

User Module

- Registration • Login • Profile Management • Upload Photos/Videos • Stories • Reels • Like • Comment • Follow • Direct Messaging

Creator Module

- Content Upload • Insights • Monetization Features

Admin Module

- User Management • Content Moderation • Reports • Analytics

SDLC Models Analysis

Waterfall: Low – Difficult to adapt to changing features.

Prototype: Medium – Helps refine UI/UX but increases cost.

Spiral: High – Excellent for managing security and scalability risks.

Agile: Very High – Supports continuous feature updates and feedback.

DevOps: Very High – Enables Continuous Integration and Continuous Deployment (CI/CD).

Why Agile is Best

- New features are released frequently.
- User feedback drives improvements.
- Security updates are continuous.
- Millions of active users require scalability.
- Bugs must be fixed quickly.

Agile Development Process

Requirement Gathering → Sprint Planning → System Design → Development → Testing → Deployment → User Feedback → Next Sprint

SDLC Phases

1. Requirement Analysis: User requirements, messaging, media sharing, notifications, recommendation engine (Deliverable: SRS).
2. System Design: Database, UI/UX, System Architecture, APIs (Deliverables: UML Diagrams, Database Schema).
3. Development: Authentication, Feed, Stories, Reels, Messaging, Notifications using Java/Kotlin, React, Python, Node.js and cloud databases.
4. Testing: Unit, Integration, System, Performance, Security and User Acceptance Testing.
5. Deployment: Cloud deployment, Android and iOS releases.
6. Maintenance: Bug fixes, feature enhancements, security patches and performance optimization.

Comparison

Agile and DevOps provide excellent support for changing requirements, rapid feature delivery, customer feedback and scalability, making them the most suitable models for Instagram.

Results

Instagram requires continuous updates, high availability, strong security and rapid deployment. Agile with DevOps enables continuous development, testing and deployment while maintaining high software quality.

Conclusion

Instagram is a dynamic social media platform where Agile enables flexibility, faster feature delivery and continuous user feedback. DevOps complements Agile by automating testing, deployment and maintenance, ensuring scalability, reliability and an excellent user experience.